ARUNA

[Nama Tim]
[Institusi] · 2026

SMART HOME BIODIGESTER

Leftovers become energy.

ARUNA anaerobically digests household leftovers into **biogas** for cooking, **liquid fertilizer** and **compost**, monitored by sensors and an app.

SDG 7 · SDG 11 · SDG 12 · SDG 13



01 · PROBLEM

±40%
of national waste is food
SIPSN KLHK

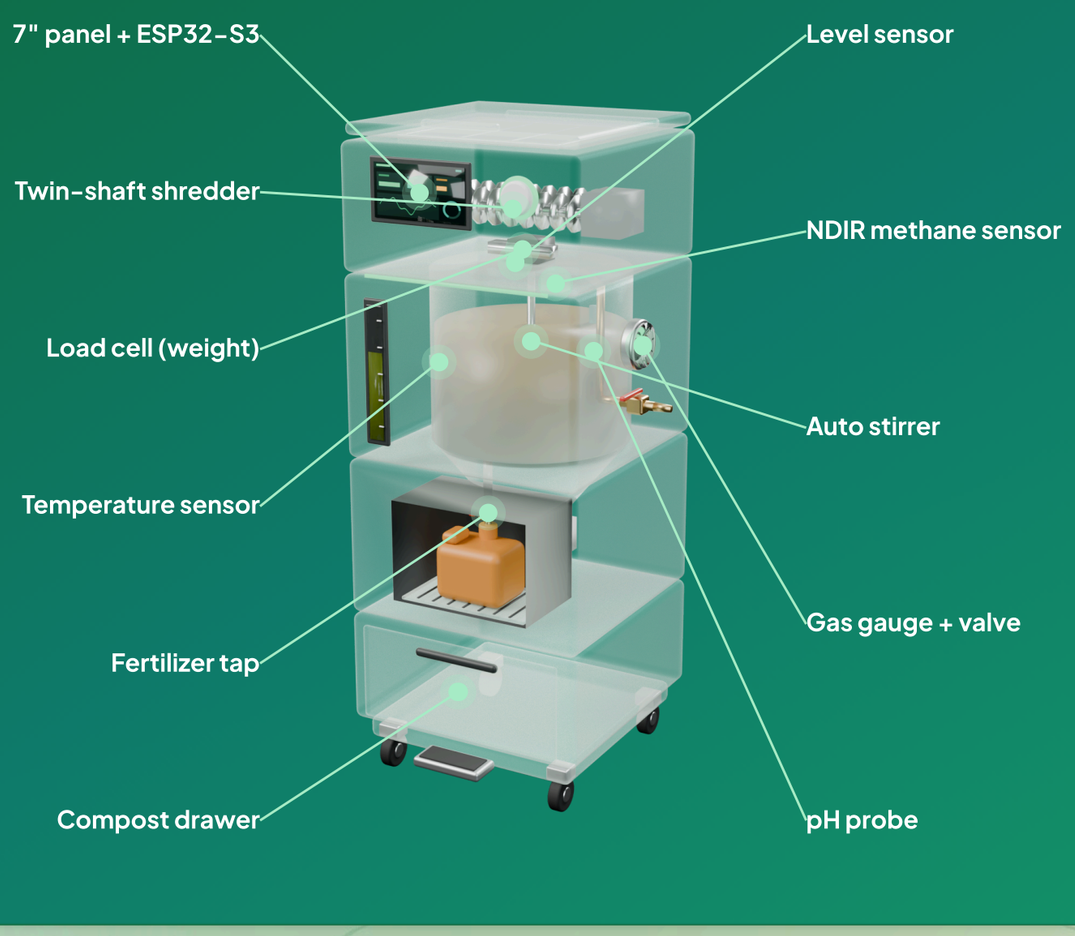
115–184 kg
food wasted / person / year
Bappenas, 2021

Rp213–551 T
economic loss per year
Bappenas, 2021

1,7 Gt CO_2e
emissions from food loss & waste 2000–2019
Bappenas, 2021

02 · HOW IT WORKS

Gravity works, sensors watch.



03 · HOW TO USE

5 steps, 2 minutes a day.



1 Put in
Open the lid, tip the leftovers.



2 Digest
Shredded & stirred automatically, ±34 days.



3 Monitor
Status & alerts on your phone.



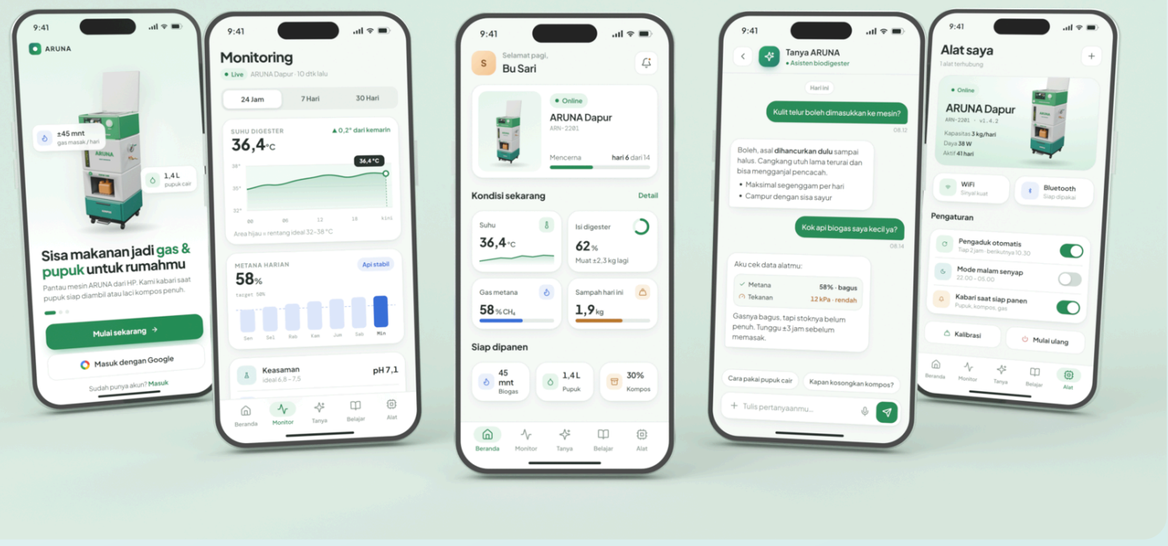
4 Fertilize
Dilute 1:10, water the plants.



5 Cook
Blue biogas flame, ±45 min/day*.

04 · APP

Everything visible on your phone.



05 · IMPACT

One kitchen, three outputs.

Cooking gas	±45 mnt*
Liquid fertilizer / cycle	±1,4 L*
Waste diverted	1,5 kg/hari*

Try the app
aruna-id.web.app

3D character: "Casual Male 3D Character – T-Pose" by sthaarpit (Sketchfab, CC-BY 4.0), re-rigged with Rigify. *Output figures are prototype design targets.